

Thursday, November 20, 2025

Data flow diagram

Got you. Let's lock in the **Data Flow Diagram (DFD)** in a way you can *copy-paste straight into your federal pack* and later turn into a visual diagram if needed.

I'll give you:

1. **Level 0 – Context DFD (text version)**
2. **Level 1 – Detailed NI-AI Data Flows**
3. **A clean narrative paragraph** you can drop under "Data Flow Diagram" in any government form.

1. Level 0 – Context DFD (Text)

Purpose: Show, at the highest level, how NI-AI interacts with users and government systems.

External Entities (E):

- **E1 – End User / Participant**

Neurodivergent individual, caregiver, operator, or staff member interacting with NI-AI.

- **E2 – Administrator / Agency Staff**

Authorized personnel who configure policies, review logs, and manage access.

- **E3 – Government / External Systems**

Secure agency networks, EMR systems, LMS platforms, security systems, or case-management tools NI-AI connects to.

Single Core Process (P0):

- **P0 – NI-AI Platform**

Includes NI Signal Capture, Emotional State Engine, Coherence Engine, EchoAuth, and Logging.

High-Level Data Flows (F):

- **F1 – Interaction Data IN**

From **E1 End User** → **P0 NI-AI Platform**

(text, voice, click patterns, timing, basic device metadata).

- **F2 – Adaptive Response OUT**

From **P0 NI-AI Platform** → **E1 End User**

(stabilized responses, prompts, recommendations, authentication decisions).

- **F3 – Admin Commands / Config**

From **E2 Admin** → **P0 NI-AI Platform**

(policy settings, access rules, thresholds, reports).

- **F4 – Logs / Reports**

From **P0 NI-AI Platform** → **E2 Admin / Agency**

(audit logs, risk alerts, usage summaries).

- **F5 – Secure System Integration Data**

Between **P0 NI-AI Platform** ↔ **E3 Government / External Systems**

(authentication results, session tokens, event triggers, non-PII metrics).

You can describe the Level 0 diagram like this:

Level 0 Context DFD: End Users and Admins send interaction data and configuration instructions to the NI-AI Platform, which processes this information and returns adaptive responses, authentication results, and audit logs, while securely integrating with existing government/enterprise systems.

2. Level 1 – NI-AI Core Data Flow (Detailed Text)

Here we break the NI-AI Platform into main processes (P), data stores (D), and flows (F).

External Entities (same as above)

- **E1 – End User / Participant**
 - **E2 – Administrator / Agency Staff**
 - **E3 – Government / External Systems**
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Processes (P)

- **P1 – Client Interface Layer (Portal / App / Web)**

Receives user input (text, voice, clicks), enforces TLS/HTTPS, and forwards to NI-AI Core.

- **P2 – NI Signal Capture & Pre-Processing**

Normalizes text/voice, strips unnecessary identifiers, tags session with anonymous/session ID.

- **P3 – Emotional State Capture (ESC Engine)**

Analyzes rhythm, tone, and patterns → generates **Emotional State Vector** and **Drift Score**.

- **P4 – Coherence Engine (C1–C5 Layers)**

Applies NI-AI rules to keep interaction stable: context tracking, pacing, intent alignment.

- **P5 – Ori Geometry Mapping Layer**

Maps interaction patterns to internal geometric structures for prediction and stability.

- **P6 – EchoAuth Adaptive Authentication Engine**

Uses Resonance Match Score (RMS), behavioral patterns, and device signature to make **Allow / Challenge / Deny** decisions.

- **P7 – Response Generation & Stabilization**

Builds the final response, adjusted to user's emotional state, clarity level, and sensory load.

- **P8 – Audit, Logging & Monitoring Layer**

Records security events, high-level session metadata, and risk signals (no raw content unless required by policy).

Data Stores (D)

- **D1 – Policy & Configuration Store**

Access rules, thresholds, agency settings, and NI-AI operating modes.

- **D2 – Authentication & Identity Store**

Encrypted user IDs, RMS profiles, and device mappings (minimal PII, per policy).

- **D3 – Telemetry & Audit Log Store**

Encrypted logs: timestamps, outcomes (success/fail), risk flags, anonymized metrics.

- **D4 – Model & Rules Store**

NI-AI Coherence rules, Ori Geometry profiles, emotional mapping parameters.

Key Data Flows (F)

You can list them like this in your doc:

- **F1 – User Input Flow**

E1 → P1 → P2

Text/voice, click events, timing, device metadata (IP, device type where allowed).

- **F2 – Pre-Processed Interaction Flow**

P2 → P3

Normalized interaction payload (IDs replaced with session or pseudonymous tokens).

- **F3 – Emotional State Analysis Flow**

P3 → P4

Emotional State Vector, Drift Score, confidence metrics.

- **F4 – Coherence Context Flow**

P4 ↔ D4

Coherence rules, session context, past interaction summaries (where allowed).

- **F5 – Ori Geometry Pattern Flow**

P4 → P5 → P4

Pattern scores, geometric mapping results to refine stability.

- **F6 – Authentication Decision Flow**

P4 → P6 ↔ D2

Includes RMS inputs, behavioral pattern hashes, and device signature checks.

- **F7 – Policy & Threshold Flow**

P4 / P6 ↔ D1

Uses agency-defined rules to decide: allow/challenge/deny, risk score.

- **F8 – Response Assembly Flow**

P4 / P7 → P1 → E1

Stabilized answers, prompts, or authentication result to the user.

- **F9 – Log & Telemetry Flow**

P4 / P6 / P7 → P8 → D3

Writes audit trail: event type, timestamp, outcome, risk level, and system actions.

- **F10 – Government System Integration Flow**

P6 / P7 ↔ E3

Sends minimal necessary information (e.g., "user authenticated", risk flag, session token) to agency systems.

Trust / Security Boundaries

You can explicitly state:

- **Boundary B1 – Client vs. NI-AI Cloud/Gov Environment**

All traffic between **P1** and the NI-AI Core (P2–P8) is protected via **TLS 1.2+**, with mutual auth where required.

- **Boundary B2 – NI-AI Processing vs. Data Stores**

Access to **D1–D4** is controlled via **role-based access control (RBAC)**, encryption at rest, and audited access.

- **Boundary B3 – NI-AI vs. External Government Systems**

Integrations with **E3** are through hardened APIs, VPNs, or private links, with least-privilege data sharing.

3. Narrative Text Block for “Data Flow Diagram” Section

You can copy-paste this directly into a **“Data Flow Diagram”** or **“Data Flow Description”** field in a federal form:

NI-AI Data Flow Overview

End users and authorized staff access the NI-AI system through a secure client interface (web or application portal) protected by TLS. User input (text, voice, clicks, timing, and limited device metadata) is sent to the NI-AI Signal Capture component, where it is normalized and associated with a session identifier.

The Emotional State Capture (ESC) engine analyzes interaction patterns to compute an emotional state vector and drift score, which are passed into the NI-AI Coherence Engine. The Coherence Engine consults internal rules and prior session context to maintain stable, aligned responses and forwards stabilized interaction data to the Ori Geometry layer for additional pattern mapping.

The EchoAuth adaptive authentication engine combines resonance-based scores, behavioral

micro-patterns, and device signatures with agency-defined policies to determine authentication outcomes (allow, challenge, or deny). Decisions and thresholds are stored in an encrypted policy and identity store.

Final responses are generated and adjusted to the user's emotional and cognitive state (e.g., pacing, tone, complexity), then returned to the user via the secure client interface.

Throughout this process, security-relevant events, risk signals, and outcomes are written to an encrypted audit log store. Where required, NI-AI transmits minimal authentication and risk data to existing government systems via secure APIs or private network connections. No raw emotional content or full transcript is shared outside NI-AI unless explicitly authorized by policy.

If you want, the **next step** in this compliance block would be:

- **Proceed — Generate Privacy Impact Assessment (PIA Outline)**

But we can also stop and tighten what we already have first.